Sayak Chatterjee

Department of Physics
University of Massachusetts, Amherst.
690 N Pleasant Street, MA 01003

Phone: +1(413) 4720-723
Email: sayakchatter@umass.edu
sayakchatterjee896@gmail.com

PROFESSIONAL SUMMARY

Experimental physicist specializing in precision measurements, detector development such as Gas Electron Multipliers, Cherenkov, Scintillator, Resistive plate chamber and straw-tube, and high-rate data acquisition systems. Expertise in Monte Carlo simulations, GEANT4-based modeling, and experimental data analysis. Proven track record in developing advanced instrumentation, optimizing detector performance, and collaborating within international research teams. Passionate about contributing to nuclear and particle physics research at the forefront of discovery.

EDUCATION

Bose Institute, University of Calcutta, Kolkata, India **2018-2023**

PhD in experimental high energy nuclear physics

Thesis title: Performance studies of Gas Electron Multiplier detector for the Muon Chamber of high rate CBM experiment at FAIR

Bose Institute, University of Calcutta, Kolkata, India **2016-2018**

Master of Science (M.Sc.) in Physics; **Grade:** A^+

Master's dissertation title: Study of characteristics of triple GEM detector for CBM experiment

St. Paul's CM College, University of Calcutta, Kolkata, India **2013-2016**

Bachelor of Science (B.Sc.): Physics (Hons.); **Grade:** First Class

Rank: 2nd in B.Sc. Physics in the University of Calcutta

RESEARCH EXPERIENCE

University of Massachusetts, Amherst, Massachusetts, USA **2022-Present**

Post Doctoral Research Scholar in the Department of Physics

Project: Measurement of Lepton-Lepton Electroweak Reaction (MOLLER) at Jefferson Lab, Virginia, USA

- Characterization of Cherenkov detectors in the beamline and with cosmic muons
- Gas Electron Multiplier based cosmic muon tracking setup to characterize the Cherenkov detectors
- Implementing and testing of VME, MPD/APV25, fADC, QDC based readout systems
- Testing of Cherenkov detectors in the beamline at MAMI accelerator facility, Mainz, Germany
- Contributing to detector hardware development, including calibration, cosmic muon testing, and flux characterization
- Offline data analysis framework development using CERN-ROOT framework
- Testing of the final Cherenkov production modules and mechanical assemblies at the college of William & Mary
- Developing GEANT-4 based simulation to optimize the apparatus design
- Mentoring graduate (2) and undergraduate (6) students in the lab
- Presenting research findings at Jefferson lab collaboration meetings and at international conferences

Bose Institute, Kolkata, India **2018-2022**

Researcher in the Department of Physics (PhD)

Project: Compressed Baryonic Matter (CBM) experiment at FAIR, Darmstadt, Germany

- Designed and built Gas Electron Multiplier (GEM) detectors for the Muon Chamber in the CBM experiment at FAIR
- Investigated long-term stability, charging-up effects, and spark probability of triple GEM detectors under high radiation conditions
- Conducted Monte Carlo-based simulations using GEANT4 for di-muon detection efficiency and physics performance studies
- Participated in beam test campaign at GSI to characterize GEM detector performance
- Phenomenological studies on Cold Nuclear Matter (CNM) effects on J/ψ production at FAIR and SPS energies
- Characterized Resistive Plate Chamber (RPC), Straw-tube detector for CBM experiment and Scintillator for cosmic muon studies

TECHNICAL SKILLS

Experimental Techniques & Detector Systems

- Gas Electron Multipliers (GEM), Cherenkov Detectors, Resistive Plate Chambers (RPC), Straw Tube Detectors, Plastic Scintillators
- PMT & detector Calibration, X-ray and Cosmic Ray Testing, High-Rate Detector Characterization

Data Acquisition & Electronics

- Handling of VME, VTP, MPD/APV25, fADC, QDC-based readouts and standard NIM electronics

Simulation & Data Analysis

- **Monte Carlo Simulations**: GEANT4, FLUKA, UrQMD, PLUTO, Garfield
- **Data Analysis**: Python, C++, ROOT (CERN)

Miscellaneous software: Autodesk, Inventor, JT2Go

AWARDS & RECOGNITION

- Nominated by the MOLLER collaboration to present the MOLLER overview in Jefferson Lab Hall A collaboration meeting, Jefferson Lab, Virginia, USA, January 2024
- Received the award for best oral presentation in the Students' Symposium "Recent Trends in Natural Sciences, 2022" at Bose Institute, Kolkata, November 2022
- Received the Young Research Grant to attend the 15th Pisa Meeting on Advanced Detector, La Biodola, Isola d'Elba, Italy, May 2022
- Received the Ernest Rutherford best researcher award 2022 (January) by the International Multidisciplinary Research Foundation (IMRF)
- Selected as the speaker for the Highlight talk on CBM-MuCh simulation at the plenary session of 37th CBM Collaboration Meeting (Online), 5th March 2021
- Best poster award at Workshop on Dynamics of QCD Matter, National Institute of Science Education and Research (NISER), Odisha, India, August 2019
- Ph.D. research fellowship, Bose Institute (2018 - 2022), Kolkata, India

SELECTED PUBLICATIONS

- S. Chatterjee on behalf of the MOLLER Collaboration "Characterization of Cherenkov detectors for the MOLLER experiment" NIMA 1070 (2025) 170057
- S. Chatterjee *et al*., "Visual investigation of possible degradation in GEM foil under test" NIMA 1045 (2023) 167573

- S. Chatterjee *et al*., "Effect of relative humidity on the long-term operation of a single mask triple GEM chamber" NIMA 1046 (2023) 167747
- S. Chatterjee *et al*., "Charging-up effect and uniformity study of a single mask triple GEM detector" NIMA 1049 (2023) 168110
- S. Chatterjee *et al*., "J/ψ yields in low energy nuclear collisions at SPS and FAIR: A baseline estimation" Nuclear Physics A 1029 (2023) 122554
- S. Chatterjee *et al*., "Observation on the bias current variation of a single mask triple GEM chamber" JINST 18 C05002
- S. Chatterjee *et al*., "Study of charging-up effect for a single mask triple GEM detector" NIMA 1014 (2021) 165749
- S. Chatterjee *et al*., "Spark probability measurement of a single mask triple GEM detector" NIMA 977 (2020) 164334
- S. Chatterjee *et al*., "Study of charging up effect in a triple GEM detector" Journal of Instrumentation 15 (2020) T09011
- S. Chatterjee *et al*., "Study of uniformity of characteristics over the surface for triple GEM detector" NIMA 936 (2019) 491

OTHER PUBLICATIONS

- **Journal publication:** 5, **Conference proceedings:** 16, **Collaboration notes:** 13

ORAL/POSTER PRESENTATION & CONFERENCES

Oral presentation: 23, **Poster presentation:** 6, **Conferences:** 7

PROFESSIONAL & OTHER SERVICES

- Organizer & committee member of the UMass Amherst Postdoc symposium 2025
- Committee member of the UMass International Faculty & Stuff Network group, 2024-25
- Editorial board member at Radiation Science and Technology, Science Publishing Group
- Reviewer in Measurement, Nuclear Instrumentation and Methods in Physics Research A
- Judge for the Senior School Science Exhibition at La Martiniere for Boys, Kolkata, India
- Volunteer for organizing 34th CBM Collaboration meeting, at Bose Institute, Kolkata, India
- Volunteer for FAIR project at Vigyan Samagam in Mumbai, Bengaluru and Kolkata, India
- Quiz master at the FAIR week of Vigyan Samagam in Mumbai, Bengaluru and Kolkata, India
- Target Operator (TO) for the SBS Gen-II experiment at Jlab Hall A